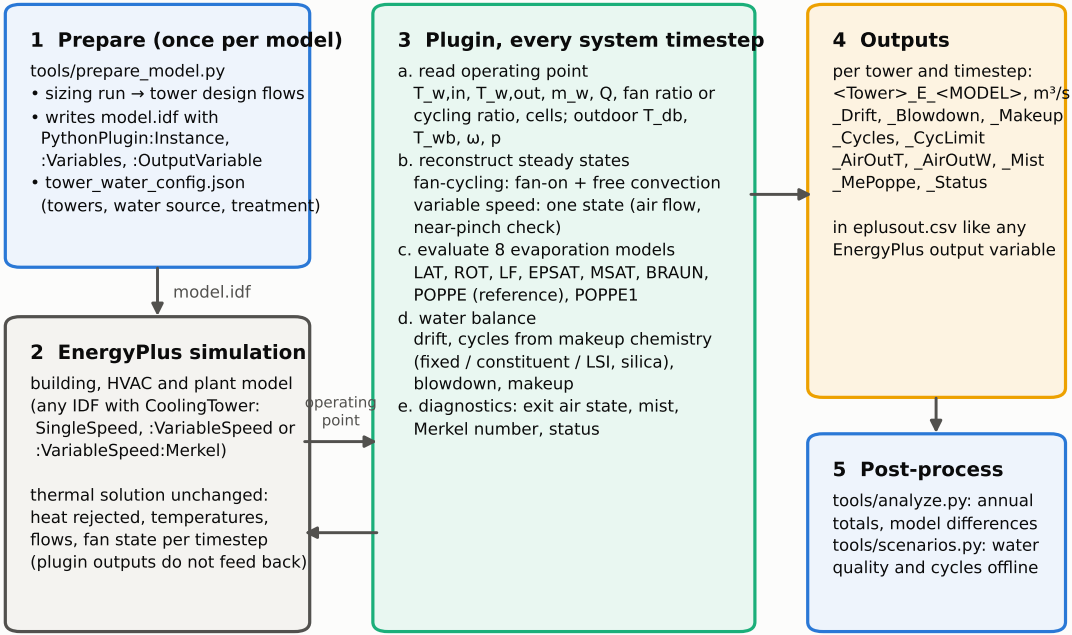
1 Prepare (once per model)
tools/prepare_model.py
• sizing run → tower design flows
• writes model.idf with PythonPlugin:Instance, :Variables, :OutputVariable
• tower_water_config.json (towers, water source, treatment)
model.idf
2 EnergyPlus simulation
building, HVAC and plant model (any IDF with CoolingTower: SingleSpeed, :VariableSpeed or :VariableSpeed:Merkel)
thermal solution unchanged: heat rejected, temperatures, flows, fan state per timestep (plugin outputs do not feed back)
operating point
3 Plugin, every system timestep
a. read operating point
T_w,in, T_w,out, m_w, Q, fan ratio or cycling ratio, cells; outdoor T_db, T_wb, ω, p
b. reconstruct steady states
fan-cycling: fan-on + free convection
variable speed: one state (air flow, near-pinch check)
c. evaluate 8 evaporation models
LAT, ROT, LF, EPSAT, MSAT, BRAUN, POPPE (reference), POPPE1
d. water balance
drift, cycles from makeup chemistry (fixed / constituent / LSI, silica), blowdown, makeup
e. diagnostics: exit air state, mist, Merkel number, status
4 Outputs
per tower and timestep:
<Tower>_E_<MODEL>, m³/s
_Drift, _Blowdown, _Makeup
_Cycles, _CycLimit
_AirOutT, _AirOutW, _Mist
_MePoppe, _Status
in eplusout.csv like any EnergyPlus output variable
5 Post-process
tools/analyze.py: annual totals, model differences
tools/scenarios.py: water quality and cycles offline